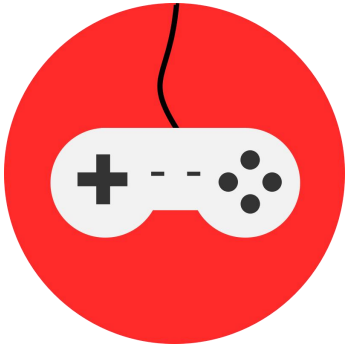
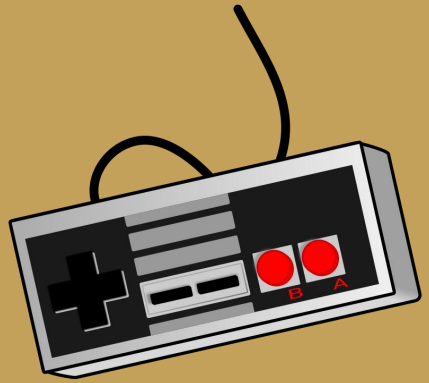
Video Games

Bryce Du, Bryce Kratzer, Sunny Jin

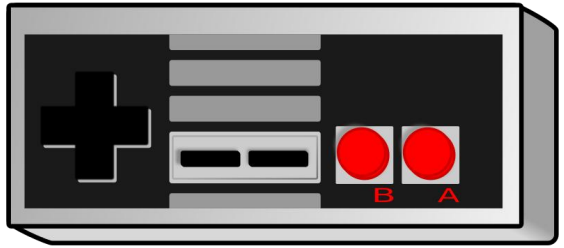


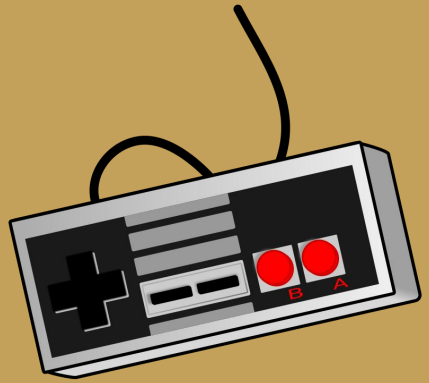


1. Intro

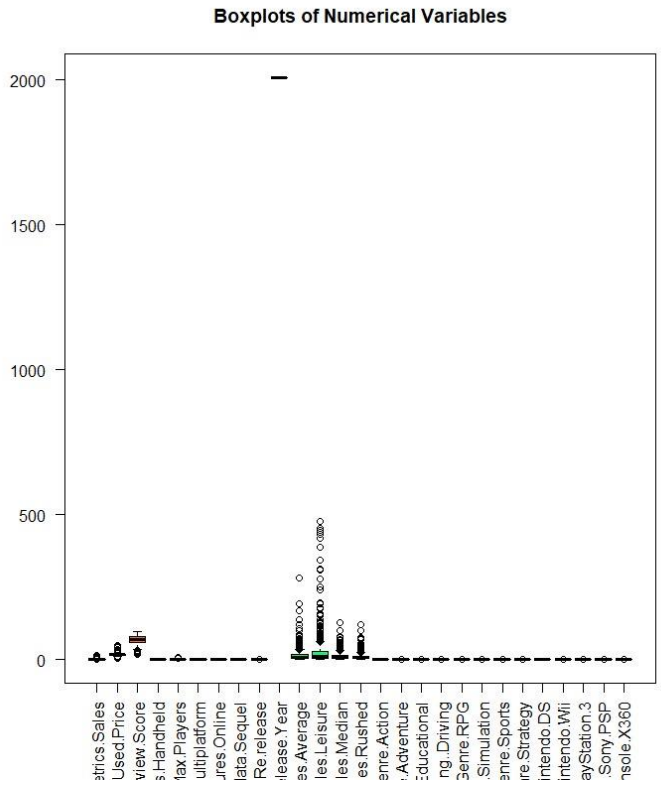
Research Rationale

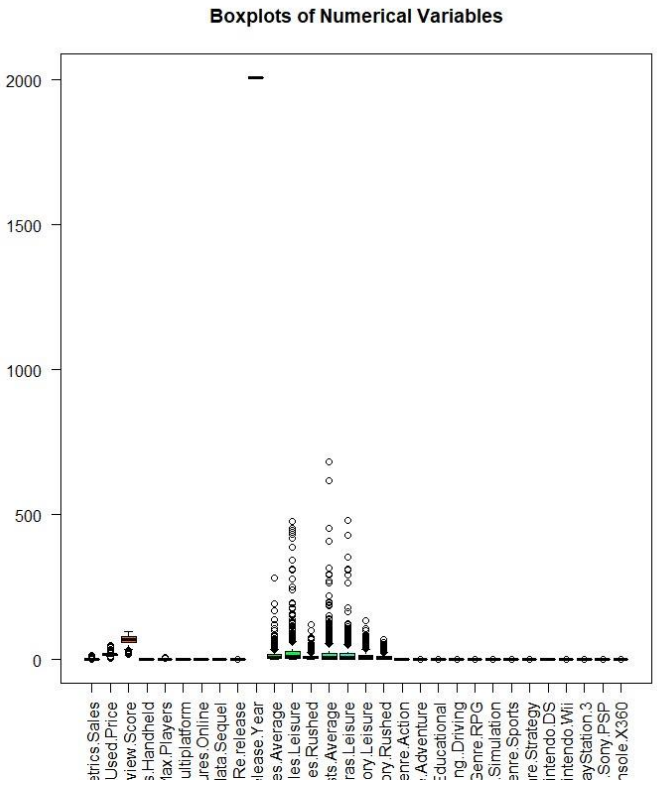
- Research question: What makes a blockbuster video game?
  - Maximize sales/profits



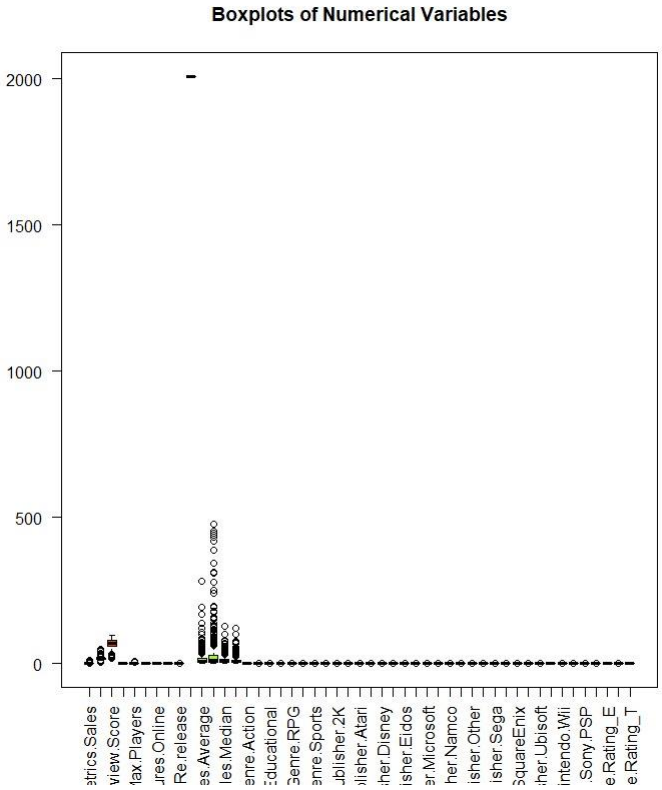


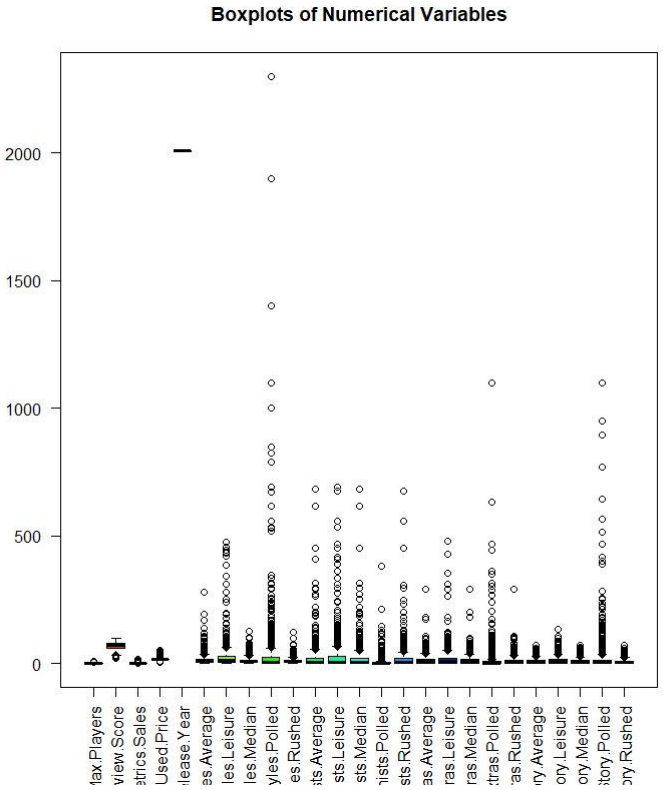
2. Data Set Description





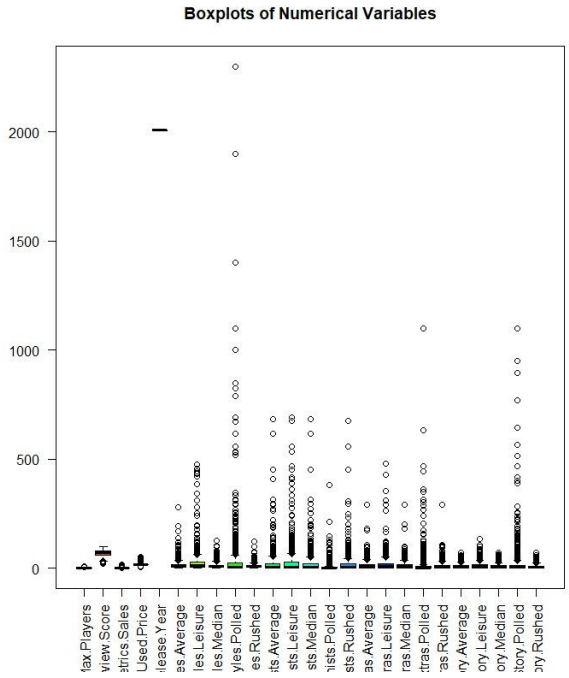
Numerical Variable Boxplots





Numerical Variable Boxplots

Full Model


Boxplots of Numerical Variables
2000
1500
1000
500
0
fax.Players
view.Score
atrics.Sales
Used.Price
lease.Year
ss.Average
les.Leisure
les.Median
yles.Polled
es.Rushed
ts.Average
sts.Leisure
sts.Median
ists.Polled
sts.Rushed
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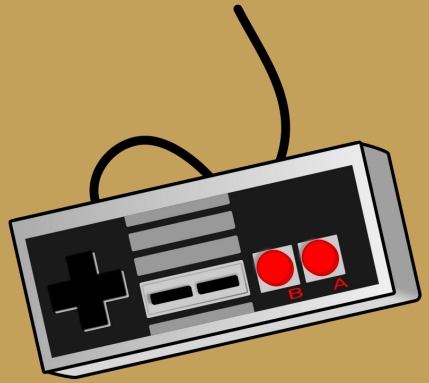
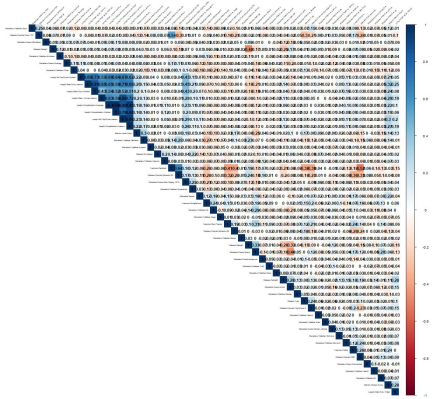


3. Building Model and Model Validation

Cleaning the data

Title	Metadata.Publishers	Release.Rating	Features.Handheld?
Pokmon Dash	Nintendo	T	TRUE
Call of Duty 2	Activision,Konami	E	TRUE
Dynasty Warriors		E	TRUE

Title	Metadata.Publishers	Release.Rating	Features.Handheld?
Pokmon Dash	Nintendo	E	TRUE
Call of Duty 2	Activision	T	FALSE
Dynasty Warriors	Other	T	TRUE



- Filled Empty Cells
- Removed Erroneous Publishers
- Corrected boolean Statements
- Corrected game ratings
- Checked for multicollinearity

Feature Selection

-Created dummy variables for categorical variables

- Utilizing LASSO for feature selection

- Iteratively adding interactions between variables

Final Model

- 57 variables down from 65

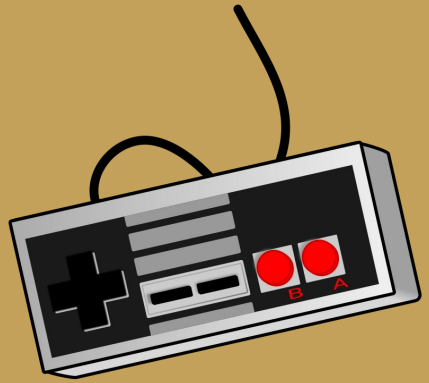
-Out of the different Genres, Action games are most likely to sell better, and Strategy games are most likely to sell worse.

-Nintendo is the most influential publisher (the only publisher with absolute coefficients value > 0.1)

-Interaction terms: Used.Price:Review.Score has a significant impact; so do Used.Price:Review.Score:PlayStyles.Leisure (largest coefficient)

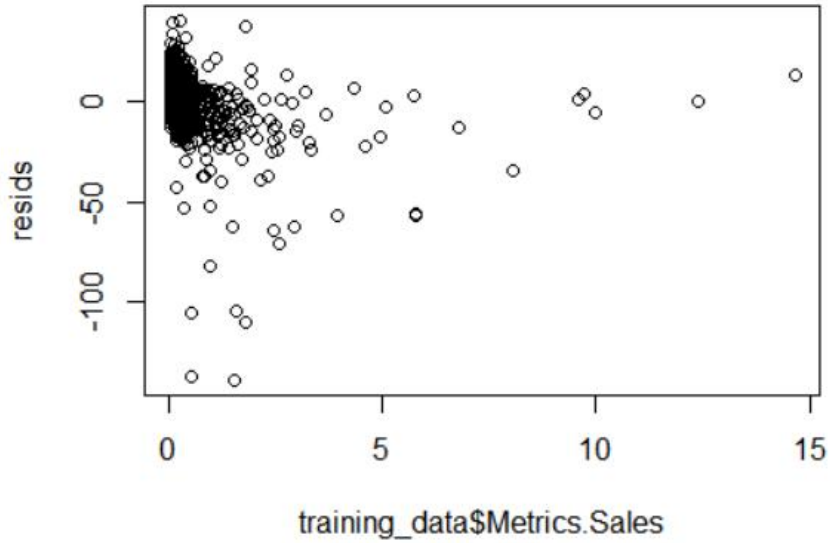
- Price & Review Score has a very high impact to the game price, and even higher when the game has a Leisure PlayStyle.

(Intercept)	0.451371158
Metrics.Used.Price	.
Metrics.Review.Score	0.079359433
Features.Handheld	.
Features.Max.Players	0.031671636
Features.Multiplatform	0.009880373
Features.Online	0.004588842
Metadata.Sequel	0.004458508
Release.Re.release	-0.01974893
Release.Year	.
Length.All.PlayStyles.Average	-0.033580094
Length.All.PlayStyles.Leisure	.
Length.All.PlayStyles.Median	.
Length.All.PlayStyles.Rushed	-0.038350341
Metadata.Genre.Action	0.036460545
Metadata.Genre.Adventure	-0.034304355
Metadata.Genre.Educational	0.003908566
Metadata.Genre.Racing..Driving	.
Metadata.Genre.RPG	-0.016467072
Metadata.Genre.Simulation	-0.005512115
Metadata.Genre.Sports	-0.001212116
Metadata.Genre.Strategy	-0.065301827
Metadata.Publisher.2K	.
Metadata.Publisher.Activision	0.000543592
Metadata.Publisher.Atari	.
Metadata.Publisher.Capcom	.
Metadata.Publisher.Disney	.
Metadata.Publisher.EA	0.021560334
Metadata.Publisher.Eidos	-0.003213285
Metadata.Publisher.Konami	-0.007583416
Metadata.Publisher.Microsoft	0.02779703
Metadata.Publisher.Midway	.
Metadata.Publisher.Namco	-0.032642589
Metadata.Publisher.Nintendo	0.118304157
Metadata.Publisher.Other	-0.003675645
Metadata.Publisher.Rockstar	0.057588216
Metadata.Publisher.Sega	0.002499996
Metadata.Publisher.Sony	0.017911192
Metadata.Publisher.SquareEnix	-0.010291641
Metadata.Publisher.THQ	.
Metadata.Publisher.Ubisoft	.
Release.Console.Nintendo.DS	.
Release.Console.Nintendo.Wii	0.011130303
Release.Console.PlayStation.3	-0.025268721
Release.Console.Sony.PSP	-0.001464916
Release.Console.X360	0.038702969
Release.Rating_E	.
Release.Rating_M	0.002166164
Release.Rating_T	-0.04995331
Metadata.Genre.Action:Metadata.Genre.RPG	-0.027431259
Metadata.Genre.Action:Metadata.Genre.Strategy	.
Metadata.Genre.Action:Metadata.Genre.Sports	.
Metadata.Genre.Action:Metadata.Publisher.Activision	0.085873113
Metadata.Genre.Action:Metadata.Publisher.Microsoft	.
Metadata.Genre.Action:Metadata.Publisher.Rockstar	0.037840914
Metrics.Used.Price:Length.All.PlayStyles.Average	.
Metrics.Used.Price:Metrics.Review.Score	0.133711444
Metrics.Used.Price:Metrics.Review.Score:Length.All.PlayStyles.Leisure	0.311411162



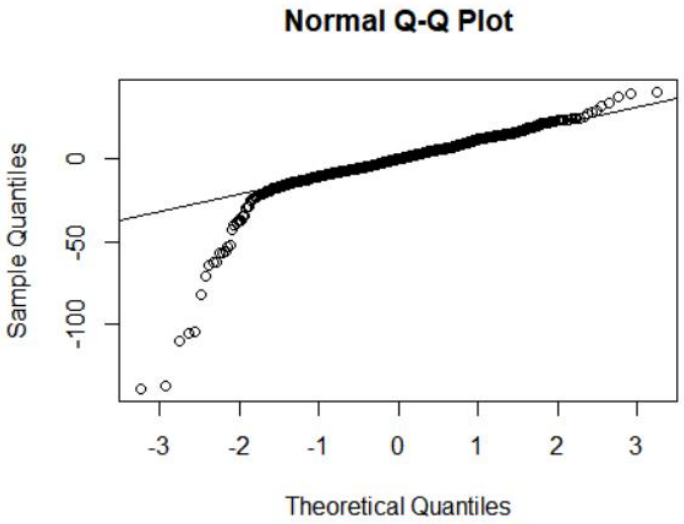
4. Model Diagonostics

Check Linearity Assumption - Residual Plot



No visible non-linear trend.

Check Normality Assumption - Normal QQ Plot




```
shapiro.test(resids)

##
##  Shapiro-Wilk normality test
##
## data:  resids
## W = 0.78487, p-value < 2.2e-16
```

Equal Variance of Errors Assumption - BF test And Remedial Measure - Box Cox Transformation

```
names(BF) <- c("t", "p-value")
BF
```

```
##           t     p-value
## -2.57508070  0.01020297
```

```
#Remedial Measures: Box-Cox
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```
lambda <- BoxCox.lambda(training_data$Metrics.Sales, lower = -2, upper = 2)
lambda
```

```
## [1] -0.2380335
```

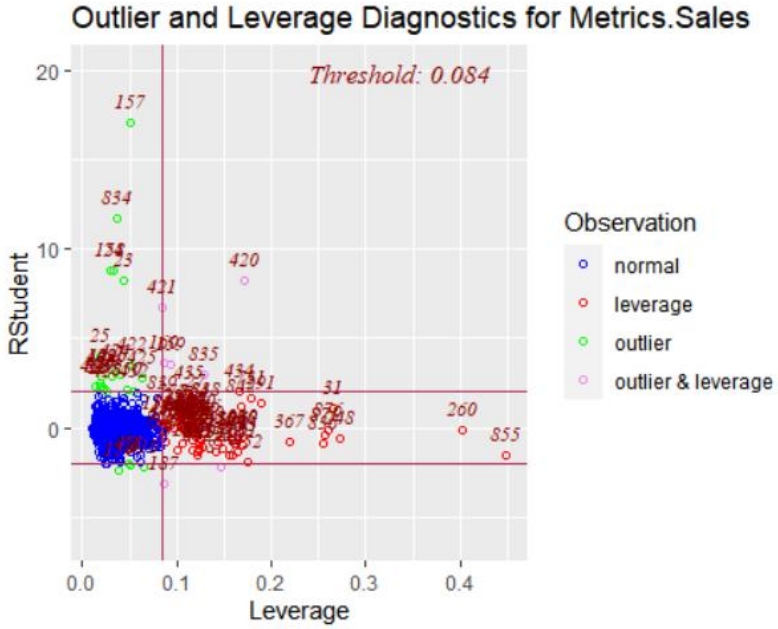
BF test result suggest that Equal Variance of Errors assumption is violated.

Remedy: Box-Cox Transformation, λ = -0.2380335

Transformed Model, and Outliers

```
c(mse,r_squared, adj_r_squared)
```

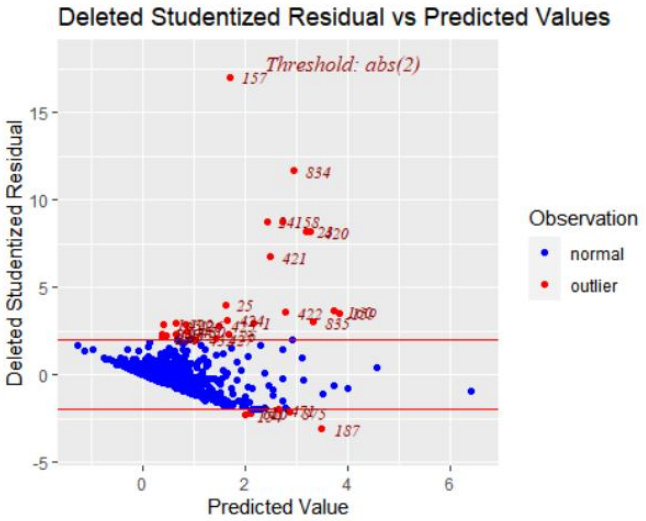
```
## [1] 0.7277747 0.3636303 0.3310384
```

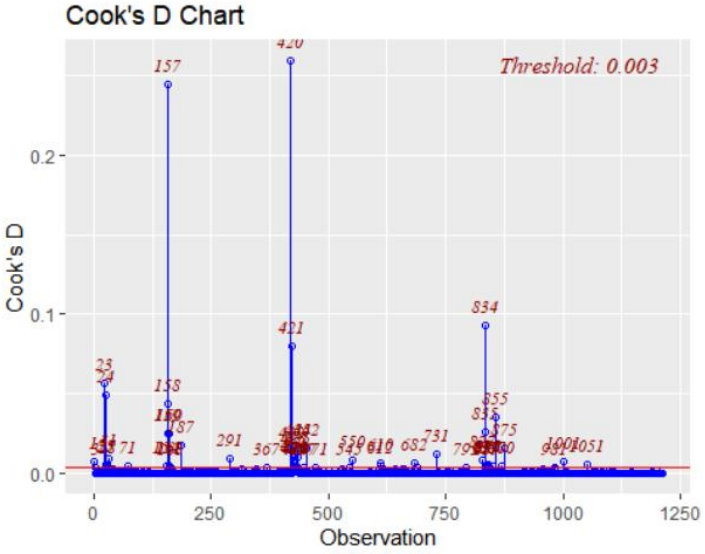


Adjusted R^2 = 0.33 is still far from ideal.

Outstanding outliers from Outlier & Leverage Diagnostics graph: # 23, 24, 134, 157, 420, 421, 834 (More on them on the next pages…)

Outliers - Studentized Residuals and Cook's Distance





The 7 Most Influential Outliers

```
## [1] "Game Name"                    "Studentized Residual"    "Cook's D"

##                [,1]                                          [,2]
               [,3]
## 23  "Mario Kart DS"                "8.19888651544466"                    "0.0566198809176657"
## 24  "Nintendogs"                   "8.76562044497503"                    "0.0496393598907225"
## 157 "Wii Play"                     "17.049159381426"                     "0.244477081793348"
## 158 "New Super Mario Bros."        "8.78652529546772"                    "0.043839456389043"
## 420 "Wii Fit"                      "8.19747354735298"                    "0.259131742623072"
## 421 "Halo 3"                       "6.76469951294258"                    "0.079813494272085"
## 834 "Mario Kart Wii"               "11.6607745146173"                    "0.0926492483962602"
```

Note that 6 of them are published by Nintendo, and 1 by Microsoft.

These outliers would be excluded from the dataset to produce the final model.

R^2 and R_{adj}^2

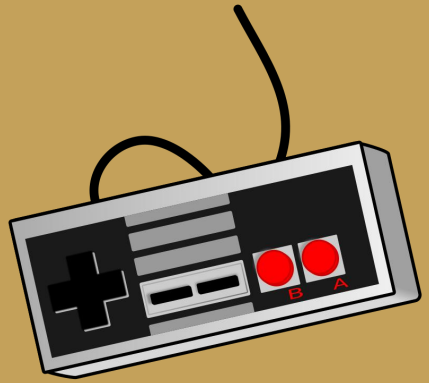
MSE, R^2 and R_{adj}^2 of the model including outliers:

[1] 0.7277747 0.3636303 0.3310384

MSE, R^2 and R_{adj}^2 of the model excluding outliers:

[1] 0.2824093 0.4641607 0.4365498

Significant increase in R_{adj}^2 ! (By 10%)



5. Conclusion

Concluding Remarks

- Action games are most likely to sell better, and Strategy games are most likely to sell worse.
- Nintendo is the most influential publisher (supported by Nintendo being vast majority of outliers & high coefficient for publish=Nintendo)
- Price & Review Score has a very high impact to the game price, and even higher when the game has a Leisure PlayStyle.